A2 Medallion architecture

Bronze, silver and gold contracts.

A2-L1 Bronze: keep every delivery exactly as received

The strong version keeps every byte the supplier sent, labels every row with its origin, and reads the file once.

The four rules of the bronze contract	Append only. As received. Audit columns. Grain: respondent per wave per delivery file.
The four bronze audit columns	ingestion_id, source_file, ingestion_timestamp, delivery_version.
Why no inferSchema in bronze?	It guesses types and can destroy values, such as the leading zeros in 00123.
Rows written, without a second read	DeltaTable.forName(spark, t).history(1) and operationMetrics["numOutputRows"].

A2-L2 Silver: canonical names, checked types, one row per respondent per wave

The strong version fails early and clearly on a changed delivery, and turns bad values into a number you can check instead of a crash or a silent NULL.

```
F.col("final_weight").try_cast("double")
```

What does withColumnRenamed do with a missing column?	Nothing, silently. Check the source columns yourself and raise one error listing every missing one.
cast on Spark 4.0 (Runtime 17.3 LTS) with a malformed value	Raises CAST_INVALID_INPUT, because ANSI mode is on by default.
Safe cast for silver	F.col(c).try_cast(t), then count the new NULLs and check them against a threshold.
Silver dedup key	The full grain: respondent_id and wave_id.

A2-L3 Gold: state the grain, group by all of it, guard every ratio

The strong version cannot merge two trackers, reports a true zero for zero reach, and never divides by nothing.

```
COMMENT ON TABLE research_gold.nat_metrics IS 'Grain: tracker x wave x age_group x gender x region';
ALTER TABLE research_gold.nat_metrics SET TBLPROPERTIES ('grain' = 'tracker x wave x age_group x gender x region');
```

Where do you record a gold table's grain?	In the function docstring and on the table: COMMENT ON TABLE or a custom property such as 'grain'.
sum over no matching rows	Returns NULL, not 0. Wrap it: F.coalesce(F.sum(...), F.lit(0)).
Safe ratio in PySpark	F.when(base > 0, part / base). NULL when the base is 0.
Why is SET TBLPROPERTIES ('owner' = ...) wrong?	owner is reserved. Spark 4.0 raises UNSUPPORTED_FEATURE.SET_TABLE_PROPERTY. Use ALTER TABLE ... OWNER TO.